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Inventor(s)	LIAO; Yu-Huai et al.

CAMERA DEVICE AND LONG TUNING RANGE CAMERA ASSEMBLY

Abstract

The present disclosure provides a camera device and a long tuning range camera assembly. The camera device comprises a frame body, a circuit component, a carrying component, and a magnetic conductive sheet. The frame body comprises an accommodating space. The circuit component comprises a circuit board, a magnetic sensing element, and a coil. The carrying component comprises a main body, a driving magnet, and a sensing magnet. Wherein the magnetic sensing element is configured to magnetically sense the sensing magnet to generate a specific signal, which comprises an angle between the sensing magnet and the magnetic sensing element or/and a position of the main body relative to the frame body. The long tuning range camera assembly comprises two sets of camera devices.

Inventors: **LIAO; Yu-Huai (Taipei City, TW), WU; Fu-Yuan (Taipei City, TW), HSU; Shang-Yu (Taipei City, TW)**

Applicant: **Lanto Electronic Limited** (Kunshan City, CN)

Family ID: **80754959**

Assignee: **Lanto Electronic Limited (Kunshan City, CN)**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of U.S. patent application Ser. No. 18/074,908, filed on Dec. 5, 2022, which claims the priority benefit of Chinese Patent Application Serial Number 202111511468.3, filed on Dec. 6, 2021, the full disclosure of which is incorporated herein by reference.

BACKGROUND
Technical Field
[0002] The present disclosure relates to the technical field of camera assembly, particularly to a camera device and a long tuning range camera assembly.
Related Art
[0003] Conventional voice coil motors comprise a sensing magnet and a Hall device to obtain distance information of lens movement. With the increase of various cameras module, various requirements for lens movement have been rising. Under the requirements for the long tuning range voice coil motor (VCM), it is necessary to arrange multiple Hall devices to sense position information and to obtain the position information from the long tuning range lenses Through the calculation of complex superposition signal. The above configuration needs to arrange multiple of Hall devices to correspond to sensing magnets, which would not only increase material costs but would also enlarge the overall size of the device and of which the assembly difficulty is increased.
SUMMARY
[0004] The embodiments of the present disclosure provide a camera device and a long tuning range camera assembly tended to solve the problem of the poor sensing performance to the lens position, the over cost, oversize of camera devices and long tuning range camera assembly through a combining design of magnetic sensors, sensing magnets, and the magnetic conductive sheets.
[0005] The present disclosure provides a camera device, comprising a frame body, a circuit component, a carrying component, and a magnetic conductive sheet. The frame body forms an accommodating space. The circuit component comprises a circuit board, a magnetic sensing element, and a coil. The magnetic sensing element and the coil are disposed on the circuit board. The circuit board is disposed on the frame body. The magnetic sensing element and the coil are disposed at one side of the frame body. The carrying component comprises a main body, a driving magnet, and a sensing magnet. The main body is disposed in the accommodating space and movable in the space. The sensing magnet and the driving magnet are disposed at one side of the main body. The sensing magnet corresponds to the magnetic sensing element. The driving magnet corresponds to the coil. The magnetic conductive sheet is disposed at the frame body. The magnetic conductive sheet is disposed at one side of the driving magnet or/and the sensing magnet. In addition, the magnetic sensing element is configured to sense the magnetic and then output the specific signal which contains the angle and position of main body relative to the frame body.
[0006] In one embodiment, the magnetic conductive sheet comprises a body having an elongated rectangular sheet-structural configuration.
[0007] In one embodiment, the magnetic conductive sheet comprises a body and a notch disposed

at one side of the body close to the sensing magnet.
[0008] In one embodiment, the magnetic conductive sheet comprises a body and a tab disposed at one side of the body close to the sensing magnet.
[0009] In one embodiment, the frame body comprises a bottom plate and a side plate perpendicular to the bottom plate. The accommodating space is formed between the bottom plate and the side plate. The sensing magnet is disposed on bottom surface of the main body of carrying component. The driving magnet is disposed on the side surface of the main body of carrying component. The magnetic conductive sheet is disposed on the bottom plate.
[0010] In one embodiment, the magnetic conductive sheet overlaps with the driving magnet in a direction perpendicular to the bottom plate.
[0011] In one embodiment, the magnetic conductive sheet completely overlaps with the driving magnet and partially overlaps with the sensing magnet in the direction perpendicular to the bottom plate.
[0012] In one embodiment, the magnetic conductive sheet comprises a body and two tabs disposed at one side of the body close to the sensing magnet. A notch exists between the two tabs. In the direction perpendicular to the bottom plate, the body completely overlaps with the driving magnet, the two tabs do not overlap with the sensing magnet, and the sensing magnet is partially disposed in the notch.
[0013] In one embodiment, the magnetic conductive sheet comprises a body and a tab. In the direction perpendicular to the bottom plate, the body completely overlaps with the driving magnet, the tab is disposed in the middle of the body and at one side of the body close to the sensing magnet, and the tab partially overlaps with the sensing magnet.
[0014] In one embodiment, the driving magnet comprises a first driving magnet and a second driving magnet. A surface of the first driving magnet comprising a first driving magnetic pole corresponds to one end of the sensing magnet comprising a first sensing magnetic pole. A surface of the second driving magnet comprising a second driving magnetic pole corresponds to one end of the sensing magnet comprising a second sensing magnetic pole.
[0015] In one embodiment, the first driving magnetic pole and the first sensing magnetic pole have the same polarity. The second driving magnetic pole and the second sensing magnetic pole have the same polarity.
[0016] In one embodiment, the camera device further comprises a guiding component disposed between the frame body and the main body.
[0017] In one embodiment, the guiding component comprises a first guiding groove, a second guiding groove, and a guiding rod. The first guiding groove is disposed at the frame body. The second guiding groove is disposed at the main body in a manner corresponding to the first guiding groove. The guiding rod is disposed between the first guiding groove and the second guiding groove.
[0018] In one embodiment, in a moving direction of the main body, the length of the guiding rod matches the length of the second guiding groove, the length of the first guiding groove is greater than the length of the second guiding groove, and two ends of the guiding rod are configured to abut against a groove wall of two ends of the second guiding groove and to slide in the first guiding groove.
[0019] The present disclosure provides a long tuning range camera assembly, comprising a first camera device and a second camera device. The first camera device comprises the camera device according to the above aspects and a first lens component disposed at the main body of the first camera device. The second camera device comprises the camera device according to the above aspects and a second lens component disposed at the main body of the second camera device. The first lens component corresponds to the second lens component. The first lens component is aligned with the second lens component.
[0020] In the embodiments of the present disclosure, by disposing the driving magnet and the

sensing magnet on the main body carrying the lens component, the main body can be movably disposed on the frame body, and the frame body comprises a magnetic sensing element and a magnetic conductive sheet. The magnetic sensing element corresponds to the sensing magnet, and the magnetic conductive sheet corresponds to one side of the driving magnet or/and the sensing magnet. The long tuning range camera assembly is formed by the combination of a plurality of camera devices. The magnetic flux concentrating of the sensing magnet is enhanced by the magnetic conductive sheet so that the magnetic sensing element could sense the movement of the sensing magnet with higher accuracy, and the space occupied in the housing could be reduced with the total cost lowered.

[0021] It should be understood, however, that this summary may not contain all aspects and embodiments of the present disclosure, that this summary is not meant to be limiting or restrictive in any manner, and that the disclosure as disclosed herein will be understood by one of ordinary skill in the art to encompass obvious improvements and modifications thereto.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The features of the exemplary embodiments believed to be novel and the elements and/or the steps characteristic of the exemplary embodiments are set forth with particularity in the appended claims. The Figures are for illustration purposes only and are not drawn to scale. The exemplary embodiments, both as to organization and method of operation, may best be understood by reference to the detailed description which follows taken in conjunction with the accompanying drawings in which:

[0023] FIG. **1** is a perspective view of a camera device of the present disclosure;

[0024] FIG. **2** is a cross-sectional view along line A-A′ of FIG. **1**;

[0025] FIG. **3** is a cross-sectional view along line B-B′ of FIG. **1**;

[0026] FIG. **4** is an exploded view of the camera device of the present disclosure;

[0027] FIG. **5** is another exploded view of the camera device of the present disclosure;

[0028] FIG. **6** is an exploded view showing the assembly for the camera device of the present disclosure;

[0029] FIG. **7** shows the assembly of a part of the camera device of the present disclosure;

[0030] FIG. **8** is an exploded schematic view of the first embodiment of a magnetic structural combination in area C of FIG. **7**;

[0031] FIG. **9** is a top schematic view of the first embodiment of the magnetic structural combination in area C of FIG. **7**;

[0032] FIG. **10** is an exploded schematic view of the second embodiment of the magnetic structural combination in area C of FIG. **7**;

[0033] FIG. **11** is a top schematic view of the second embodiment of the magnetic structural combination in area C of FIG. **7**;

[0034] FIG. **12** is an exploded schematic view of the third embodiment of the magnetic structural combination in area C of FIG. **7**;

[0035] FIG. **13** is a top schematic view of the third embodiment of the magnetic structural combination in area C of FIG. **7**;

[0036] FIG. **14** is a perspective view of a long tuning range camera assembly of the present disclosure; and

[0037] FIG. **15** is an exploded view of the long tuning range camera assembly of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0038] The present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the disclosure are shown. This present disclosure may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this present disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art.

[0039] Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but function. In the following description and in the claims, the terms "include/including" and "comprise/comprising" are used in an open-ended fashion, and thus should be interpreted as "including but not limited to". "Substantial/substantially" means, within an acceptable error range, the person skilled in the art may solve the technical problem in a certain error range to achieve the basic technical effect.

[0040] The following description is of the best-contemplated mode of carrying out the disclosure. This description is made for the purpose of illustration of the general principles of the disclosure and should not be taken in a limiting sense. The scope of the disclosure is best determined by reference to the appended claims.

[0041] Moreover, the terms "include", "contain", and any variation thereof are intended to cover a non-exclusive inclusion. Therefore, a process, method, object, or device that includes a series of elements not only includes these elements, but also includes other elements not specified expressly, or may include inherent elements of the process, method, object, or device. If no more limitations are made, an element limited by "include a/an . . . " does not exclude other same elements existing in the process, the method, the article, or the device which includes the element.

[0042] FIG. **1** is a perspective view of a camera device of the present disclosure. FIG. **2** is a cross-sectional view along line A-A′ of FIG. **1**. FIG. **3** is a cross-sectional view along line B-B′ of FIG. **1**. FIG. **4** is an exploded view of the camera device of the present disclosure. FIG. **5** is another exploded view of the camera device of the present disclosure. As shown in the figures, the present disclosure provides a camera device **1**, which comprises a frame body **11**, a circuit component **12**, a carrying component **13**, and a magnetic conductive sheet **14**. The frame body **11** comprises an accommodating space **110**. The circuit component **12** comprises a circuit board **121**, a magnetic sensing element **122**, and a coil **123**. The magnetic sensing element **122** and the coil **123** are disposed on the circuit board **121**. The circuit board **121** is disposed at the frame body **11**. The circuit board **121** is flexible. The magnetic sensing element **122** and the coil **123** are disposed at one side of the frame body **11**. The carrying component **13** comprises a main body **131**, a driving magnet **132**, and a sensing magnet **133**. The main body **131** is movably disposed in the accommodating space **110**. The sensing magnet **133** and the driving magnet **132** are disposed at one side of the main body **131**. The sensing magnet **133** corresponds to the magnetic sensing element **122**. The driving magnet **132** corresponds to the coil **123**. The magnetic conductive sheet **14** is disposed at the frame body **11** and is disposed at one side of the driving magnet **132** or/and the sensing magnet **133**. Wherein the magnetic sensing element **122** is configured to magnetically sense the sensing magnet **133** to generate a specific signal, which comprises information of an angle between the sensing magnet **133** and the magnetic sensing element **122** or/and position of the main body **131** relative to the frame body **11**.

[0043] In this embodiment, The specific signal comprises the sensing signal of the magnetic sensing element **122**, which indicates that the magnetic sensing element **122** is fixed at the position of the frame body **11**, and the sensing magnet **133** is fixed at the position of the main body **131**. Wherein the main body **131** could move relative to the frame body **11**, and when the magnetic sensing element **122** senses each of the various positions of the sensing magnet **133**, a data corresponding to the above-mentioned angle or position would be generated, namely the specific signal. Since the specific signal is not limited to the data described above, users could also calculate other required data through the sensing signal of the magnetic sensing element **122**.

[0044] Referring to FIG. **4** and FIG. **5** again, in this embodiment, the frame body **11** comprises a bottom plate **111** and two side plates **112** vertically disposed at two side edges of the bottom plate **111**. An accommodating space **110** is formed between the bottom plate **111** and the two side plates **112**. The circuit board **121** of the circuit component **12** is disposed on an outer surface of the frame body **11**. The bottom plate **111** comprises a through hole **1110**, the magnetic sensing element **122** is disposed on the circuit board **121**, and the magnetic sensing element **122** is disposed in the through hole **1110** of the bottom plate **111**. The coil **123** is disposed on the circuit board **121**, an outer surface of each of the two side plates **112** of the frame body **11** comprises a coil recess **1121** in which the coil **123** is embedded. In this way, the coil **123** could be disposed between the circuit board **121** and each of the two side plates **112** of the frame body **11**.

[0045] The main body **131** is assembled and disposed in the accommodating space **110** of the frame body **11**. The camera device **1** further comprises a lens component **15** assembled to the main body **131**. The main body **131** can move back and forth in the accommodating space **110**. Besides, the camera device **1** further comprises a housing **17** assembled to the frame body **11** so that the main body **131** is limited to be in the accommodating space **110**. The sensing magnet **133** is disposed on a side surface of the main body **131** close to the bottom plate **111**. One side of the main body **131** opposite to the magnetic sensing element **122** comprises a first recess **1311** in which the sensing magnet **133** is embedded. The magnetic sensing element **122** could perform electromagnetic sensing on the sensing magnet. The driving magnet **132** is disposed on a side surface of the main body **131** close to the side plate **112**. Two sides of the main body **131** relative to the coil **123** comprise a plurality of second recesses **1312**. The driving magnet **132** is embedded in the second recess **1312**, and the magnetic field could be changed after the coil **123** is energized to generate a magnetic force on the driving magnet **132**. In this way, the driving magnet **132** could drive the main body **131** to move within the accommodating space **110**. The magnetic conductive sheet **14** is disposed on the bottom plate **111**. Wherein a bottom surface of the bottom plate **111** of the frame body **11** comprises a bottom recess **1111** in which the magnetic conductive sheet **14** is embedded. In the present disclosure, by disposing the magnetic conductive sheet **14** on one side of the driving magnet **132** or/and the sensing magnet **133**, the magnetic conductive sheet **14** could be used to enhance the concentrating of the magnetic flux of the driving magnet **132** or/and the sensing magnet **133**, enabling the magnetic sensing element **122** to sense the change of the position or distance of the sensing magnet **133** relative to the magnetic sensing element **122**.

[0046] Referring to FIG. **8**, the driving magnet **132** comprises a first driving magnet **1321** and a second driving magnet **1322**. A surface of the first driving magnet **1321** comprising a first driving magnetic pole **13211** corresponds to one end of the sensing magnet **133** comprising a first sensing magnetic pole **1331**. A surface of the second driving magnet **1322** comprising a second driving magnetic pole **13221** corresponds to another end of the sensing magnet **133** comprising a second sensing magnetic pole **1332**. Wherein the first driving magnetic pole **13211** and the first sensing magnetic pole **1331** could have the same or different polarities. The second driving magnetic pole **13221** and the second sensing magnetic pole **1332** could have the same or different polarities. Furthermore, when the first driving magnetic pole **13211** and the first sensing magnetic pole **1331** have the same polarity, the second driving magnetic pole **13221** and the second sensing magnetic pole **1332** have the same polarity, and the driving magnet **132** and the sensing magnet **133** are mutually repelling, a better magnetic fluxing effect could be provided for the magnetic sensing element **122** to sense the changes of the position or distance of the sensing magnet **133** relative to the magnetic sensing element **122**.

[0047] FIG. **6** is an exploded view showing the assembly for the camera device of the present disclosure. FIG. **7** shows the assembly of a part of the camera device of the present disclosure. As shown in the figures, the main body **131** could move back and forth in the frame body **11**. The camera device **1** further comprises a guiding component **16** disposed between the frame body **11** and the main body **131**. Wherein the guiding component **16** comprises a first guiding groove **161**, a second guiding groove **162**, and a guiding rod **163**. The first guiding groove **161** is disposed at the frame body **11**. The second guiding groove **162** is disposed at the main body **131** in a manner corresponding to the first guiding groove **161**. The guiding rod **163** is disposed between the first guiding groove **161** and the second guiding groove **162**. That is, a part of the guiding rod **163** is disposed in the first guiding groove **161**, and another part of the guiding rod **163** is disposed in the second guiding groove **162**. In a moving direction of the main body **131**, the length of the guiding rod **163** matches the length of the second guiding groove **162**, the length of the first guiding groove **161** is greater than the length of the second guiding groove **162**, and two ends of the guiding rod **163** are configured to abut against a groove wall of two ends of the second guiding groove **162** and to slide in the first guiding groove **161**. In this embodiment, when the main body **131** moves, the groove wall of the second guiding groove **162** of the main body **131** would push one end of the guiding rod **163** to enable the guiding rod **163** to move in the first guiding groove **161** until the other end of the guiding rod **163** abuts a groove wall of the first guiding groove **161**.

[0048] In this embodiment, the circuit board **121** guides electric current through the coil **123** to change the magnetic field. The magnetic field of the coil **123** forces on the driving magnet **132** so that the driving magnet **132** would drive the main body **131** to move back and forth along the Z-axis in the accommodating space **110** of the frame body **11**. The main body **131** drives the sensing magnet **133** to move back and forth along the Z-axis as well, and the magnetic sensing element **122** could sense and measure the position or distance change of the sensing magnet **133** relative to the magnetic sensing element **122**. Wherein, when the magnetic sensing element **122** senses the movement of the sensing magnet **133**, it could sense the angle between the sensing magnet **133** and the magnetic sensing element **122**, the angle is then converted to travel distance in Z-axis by calculation (eg. trigonometry). In this case, only a single magnetic sensing element **122** (eg. Tunnel Magnetoresistance Sensor, TMR Sensor) can sense the movement of the main body **131** of a long tuning range camera assembly. In this way, the position of the main body **131** relative to the frame body **11** could be adjusted

[0049] FIG. **8** and FIG. **9** are exploded and top schematic views of the first embodiment of a magnetic structural combination in area C of FIG. **7**. As shown in the figures, in this embodiment, the magnetic conductive sheet **14** comprises a body **141** having an elongated rectangular sheet-structural configuration. The magnetic conductive sheet **14** is disposed at one side of the driving magnet **132** and the sensing magnet **133**. Wherein the sensing magnet **133** is relatively disposed at one side of the magnetic sensing element **122**. Users could partially overlap the body **141** of the magnetic conductive sheet **14**, which is in the direction perpendicular to the bottom plate **111**, with the driving magnet **132** according to requirements. In this way, the magnetic conductive sheet **14** could concentrate the magnetic flux of the driving magnet **132**.

[0050] Moreover, as shown in the figures, in the direction perpendicular to the bottom plate **111**, the body **141** of the magnetic conductive sheet **14** extends in a direction toward the sensing magnet **133**. The body **141** of the magnetic conductive sheet **14** completely overlaps with the driving magnet **132**, and the body **141** of the magnetic conductive sheet **14** partially overlaps with the sensing magnet **133**. The magnetic sensing element **122** is distant from the sensing magnet **133** by a distance D**1** along the X-axis, and the magnetic sensing element **122** is distant from the sensing magnet **133** by a distance D**2** along the Y-axis. The sensing magnet **133** is displaced along the Z-axis along with the main body **131**. The magnetic sensing element **122** could sense and measure the position of the sensing magnet **133**, that is, the magnetic sensing element **122** is distant from the sensing magnet **133** by a distance D**3** along the Z-axis. In this embodiment, the magnetic conductive sheet **14** could enhance the concentrating of the magnetic flux of the driving magnet **132** and the sensing magnet **133**, which increases the accuracy of the position or distance change of the sensing magnet **133** relative to the magnetic sensing element **122** sensed and measured by the magnetic sensing element **122**.

[0051] FIG. **10** and FIG. **11** are exploded and top schematic views of the second embodiment of the magnetic structural combination in area C of FIG. **7**. As shown in the figures, the difference between this embodiment and the first embodiment lies in the shape of the magnetic conductive sheet. In this embodiment, the magnetic conductive sheet **14**A comprises a body **141**A and two tabs **143**A disposed at one side of the body **141**A close to the sensing magnet **133**, and a notch **142**A exists between the two tabs **143**A. In the direction perpendicular to the bottom plate **111**, the body **141**A completely overlaps with the driving magnet **132**, the two tabs **143**A do not overlap with the sensing magnet **133**, and the sensing magnet **133** is partially disposed in notch **142**A. In this embodiment, through the structural change of the notch **142**A, the magnetic flux concentrating of the magnetic conductive sheet **14**A would also be changed accordingly to enhance the magnetic flux concentrating of the driving magnet **132** and the sensing magnet **133**. In this embodiment, users could select the shape and structural design of the magnetic conductive sheet **14**A according to requirements, so as to adjust the magnetic sensing element **122** for sensing and measuring the sensing magnet **133**.

[0052] FIG. **12** and FIG. **13** are exploded and top schematic views of the third embodiment of the magnetic structural combination in area C of FIG. **7**. As shown in the figures, the difference between this embodiment and the first embodiment lies in the shape of the magnetic conductive sheet. In this embodiment, the magnetic conductive sheet **14**B comprises a body **141**B and a tab **143**B. In the direction perpendicular to the bottom plate **111**, the body **141**B completely overlaps with the driving magnet **132**. The tab **143**B is disposed in the middle of the body **141**B and is disposed at one side of the body **141**B close to the sensing magnet **133**. The tab **143**B extends in the direction toward the sensing magnet **133** to partially overlap with the sensing magnet **133**. The moving path of the sensing magnet **133** pasts the tab**143**B of the magnetic conductive sheet **14**B. In this embodiment, through the structural change of the tab **143**B, the magnetic conductive sheet **14**B would correspondingly change corresponding to the enhancement of the magnetic flux concentrating of the driving magnet **132** and the sensing magnet **133**. In this embodiment, users could select the shape and structural design of the magnetic conductive sheet **14**B according to requirements, so as to adjust the magnetic sensing element **122** for sensing the sensing magnet **133**.

[0053] In this embodiment, the magnetic conductive sheet **14** comprises a body **141**, and one side of the body is provided with a notch or/and a tab. Wherein the notch is disposed at one side of the body **141** close to the sensing magnet **133** or/and the tab is disposed at one side of the body **141** close to the sensing magnet **133**. The magnetic conductive sheet **14** of the above-mentioned embodiment could change the structural configuration of the body **141** through the notch or the tab, which can correspondingly change the magnetic flux of the driving magnet **132** and the sensing magnet **133**.

[0054] FIG. **14** and FIG. **15** are perspective view and exploded view of a long tuning range camera assembly of the present disclosure. As shown in the figures, this embodiment provides a long tuning range camera assembly **2**, in which the lens inside has a relatively long traveling path, about 4 mm, which is for example a zoom lens. The long tuning range camera assembly **2** comprises a first camera device **21** and a second camera device **22**. The first camera device **21** comprises the aforementioned camera device **1**, and the first camera device **21** comprises a first lens component **211** disposed at the main body **131** of the first camera device **21**. The second camera device **22** comprises the aforementioned camera device **1**, and the second camera device **22** comprises a second lens component **221** disposed at the main body **131** of the second camera device **22**. The first lens component **211** corresponds to the second lens component **221**, and the first lens component **211** is aligned with the second lens component **221** along an axis. The first lens component **211** is moveable relative to the second lens component along the axis. In this embodiment, in the long tuning range camera assembly **2**, by adjusting the position of the first lens component **211** of the first camera device **21** and the position of the second lens component **221** of the second camera device **22**, the focusing or zooming could be realized.

[0055] In summary, embodiments of the present disclosure provide a camera device and a long tuning range camera assembly, by disposing the driving magnet and the sensing magnet on the main body carrying the lens component, the main body can be movably disposed on the frame body, and the frame body comprises a magnetic sensing element and a magnetic conductive sheet. The magnetic sensing element corresponds to the sensing magnet, and the magnetic conductive sheet corresponds to one side of the driving magnet or/and the sensing magnet. The long tuning range camera assembly is formed by the combination of a plurality of camera devices. The magnetic flux concentrating of the sensing magnet is enhanced by the magnetic conductive sheet so that the magnetic sensing element could sense the movement of the sensing magnet with higher accuracy, and the space occupied in the housing could be reduced with the total cost lowered.
[0056] It is to be understood that the term “comprises”, “comprising”, or any other variants thereof, is intended to encompass a non-exclusive inclusion, such that a process, method, article, or device of a series of elements not only comprise those elements but further comprises other elements that are not explicitly listed, or elements that are inherent to such a process, method, article, or device. An element defined by the phrase “comprising a . . . ” does not exclude the presence of the same element in the process, method, article, or device that comprises the element.
[0057] Although the present disclosure has been explained in relation to its preferred embodiment, it does not intend to limit the present disclosure. It will be apparent to those skilled in the art having regard to this present disclosure that other modifications of the exemplary embodiments beyond those embodiments specifically described here may be made without departing from the spirit of the disclosure. Accordingly, such modifications are considered within the scope of the disclosure as limited solely by the appended claims.

Claims

1. A camera device, comprising: a frame body comprising an accommodating space; a circuit component comprising a circuit board and a coil, wherein the coil is disposed on the circuit board and at one side of the frame body, the circuit board is disposed at the frame body; a carrying component comprising a main body and a driving magnet, wherein the main body is movably disposed in the accommodating space, the driving magnet is disposed at one side of the main body, the driving magnet corresponds to the coil, and the driving magnet is configured to interact with the coil; a magnetic sensing element and a sensing magnet, wherein one of the magnetic sensing element and the sensing magnet is disposed on the circuit board, and another one of the magnetic sensing element and the sensing magnet is disposed at one side of the main body; wherein the sensing magnet corresponds to the magnetic sensing element; and a magnetic conductive sheet disposed at the frame body, wherein the magnetic conductive sheet is disposed at one side of the driving magnet or/and the sensing magnet, the magnetic sensing element is configured to magnetically sense the sensing magnet to generate a specific signal, the specific signal comprises an angle between the sensing magnet and the magnetic sensing element or/and a position of the main body relative to the frame body.
2. The camera device according to claim 1, wherein the specific signal comprises the angle between the sensing magnet, the magnetic sensing element and the position of the main body relative to the frame body.
3. The camera device according to claim 1, wherein the circuit board is disposed on an outer surface of the frame body.
4. The camera device according to claim 3, wherein the frame body further comprises a through hole, and the magnetic sensing element is disposed in the through hole.
5. The camera device according to claim 3, wherein the frame body further comprises a coil recess, and the coil is embedded in the coil recess.
6. The camera device according to claim 1, wherein the frame body further comprises a bottom

recess, and the magnetic conductive sheet disposed at the bottom recess.

7. The camera device according to claim 1, wherein the frame body comprises a bottom plate and a side plate perpendicular to the bottom plate; the accommodating space is formed between the bottom plate and the side plate; the magnetic conductive sheet is disposed on the bottom plate.

8. The camera device according to claim 7, wherein the magnetic conductive sheet overlaps with the driving magnet in a direction perpendicular to the bottom plate.

9. The camera device according to claim 7, wherein the magnetic conductive sheet completely overlaps with the driving magnet and partially overlaps with the sensing magnet in a direction perpendicular to the bottom plate.

10. The camera device according to claim 7, wherein the magnetic conductive sheet comprises a body and two tabs which are disposed at one side of the body close to the sensing magnet; a notch exists between the two tabs; in the direction perpendicular to the bottom plate, the body completely overlaps with the driving magnet, the two tabs do not overlap with the sensing magnet, and the sensing magnet is partially disposed in the notch.

11. The camera device according to claim 7, wherein the magnetic conductive sheet comprises a body and a tab; in the direction perpendicular to the bottom plate, the body completely overlaps with the driving magnet, the tab is disposed in the middle of the body and at one side of the body close to the sensing magnet, and the tab partially overlaps with the sensing magnet.

12. The camera device according to claim 7, wherein the sensing magnet is disposed on a side surface of the main body close to the bottom plate; the magnetic sensing element is disposed on the circuit board and at one side of the frame body.

13. The camera device according to claim 1, wherein the driving magnet comprises a first driving magnet and a second driving magnet; a surface of the first driving magnet comprising a first driving magnetic pole corresponds to one end of the sensing magnet comprising a first sensing magnetic pole; a surface of the second driving magnet comprising a second driving magnetic pole corresponds to one end of the sensing magnet comprising a second sensing magnetic pole.

14. The camera device according to claim 13, wherein the first driving magnetic pole and the first sensing magnetic pole have different polarities; the second driving magnetic pole and the second sensing magnetic pole have different polarities.

15. The camera device according to claim 1, further comprising a guiding component disposed between the frame body and the main body.

16. The camera device according to claim 15, wherein the guiding component comprises a first guiding groove, a second guiding groove, and a guiding rod; the first guiding groove is disposed at the frame body; the second guiding groove is disposed at the main body in a manner corresponding to the first guiding groove; the guiding rod is disposed between the first guiding groove and the second guiding groove.

17. The camera device according to claim 16, wherein in a moving direction of the main body, a length of the guiding rod matches a length of the second guiding groove, a length of the first guiding groove is greater than the length of the second guiding groove, and two ends of the guiding rod are configured to abut against a groove wall of two ends of the second guiding groove and to slide in the first guiding groove.

18. A long tuning range camera assembly, comprising: a first camera device comprising: a frame body comprising an accommodating space; a circuit component comprising a circuit board and a coil, wherein the coil is disposed on the circuit board and at one side of the frame body, the circuit board is disposed at the frame body; a carrying component comprising a main body and a driving magnet, wherein the main body is movably disposed in the accommodating space, the driving magnet is disposed at one side of the main body, the driving magnet corresponds to the coil, and the driving magnet is configured to interact with the coil; a magnetic sensing element and a sensing magnet, wherein one of the magnetic sensing element and the sensing magnet is disposed on the circuit board, and another one of the magnetic sensing element and the sensing magnet is disposed at one side of the main body; wherein the sensing magnet corresponds to the magnetic sensing element; a magnetic conductive sheet disposed at the frame body, wherein the magnetic conductive sheet is disposed at one side of the driving magnet or/and the sensing magnet, the magnetic sensing element is configured to magnetically sense the sensing magnet to generate a specific signal, the specific signal comprises an angle between the sensing magnet and the magnetic sensing element or/and a position of the main body relative to the frame body; a first lens component disposed at the main body of the first camera device; and a second camera device comprising a second lens component, the first lens component corresponding to the second lens component, the first lens component being aligned with the second lens component along an axis, and the first lens component being movable relative to the second lens component along the axis.

19. The long tuning range camera assembly according to claim 18, wherein the first camera device further comprises a guiding component disposed between the frame body and the main body.

20. The long tuning range camera assembly according to claim 19, wherein the guiding component comprises a first guiding groove, a second guiding groove, and a guiding rod; the first guiding groove is disposed at the frame body; the second guiding groove is disposed at the main body in a manner corresponding to the first guiding groove; the guiding rod is disposed between the first guiding groove and the second guiding groove.